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Proposal Review 1 : 2502380

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Agency Name:	National Science Foundation
Agency Tracking Number:	2502380
Organization:	Division Of Mathematical Sciences
NSF Program:	ALGEBRA,NUMBER THEORY,AND COM
PI/PD:	Bruce, Juliette
Application Title:	Tropicalizations of Shimura Varieties
Rating:	Multiple Rating: (Excellent/Very Good)

Review

Summary

Proposal Synopsis

The PI proposes to study the tropicalization of Shimura varieties using techniques from homological algebra, algebraic geometry, and combinatorics. The broader impacts include continu organization of several conference series, with an emphasis on inclusivity, as well as advocacy through SLMath-hosted AWM retreats.

Intellectual Merit

Strengths

Rating: Very good

Based on previous work with collaborators, the PI outlines an ambitious plan to extend results on the cohomology of Shimura varieties, which parameterize objects like elliptic curves or at varieties. Several conjectures are presented with detailed approaches. In particular, the PI breaks down 4 major directions including:

1. Developing a theory of tropicalizations of Shimura varieties.
2. Giving a modular interpretation of the tropicalizations.
3. Compute weight zero cohomologies of the associated spaces, perhaps with computer software.
4. Explore additional algebraic structures on these cohomologies.

The PI and collaborators have several publications and in-progress projects working toward the stated goals of this proposal.

The PI outlines a couple projects suitable for student advisees.

Weaknesses

Many of the technical details are omitted for readability, but that means the proposed results and their consequences are at times difficult to judge.

Some of the conjectures seem a bit speculative in full generality but the PI has a good track record for producing such results.

Broader Impacts

Strengths

Rating: Excellent

The PI has a strong record of organizing conferences and meetings, especially for women and LGBTQ+ mathematicians. PI proposes to continue this work including a satellite conference 2026 ICM, expanding the GEMS workshops for gender equity in math, and a yearly retreat at SLMath for women to promote advocacy for women in mathematics. The PI serves on the S Committee on Women in Mathematics and also plans to organize a summer REU for LGBTQ+ mathematicians. The PI expects to begin mentoring students soon.

Weaknesses

The PI does not have graduate advisees but that is understandable.

Additional Solicitation-specific Review Criteria

Strengths & Weaknesses

During their most recent funding period, the PI organized more than 10 national and international conferences. They served on the Executive Committee of AWM, as a board member for Spectra, and the SLMath Committee on Women in Mathematics. The PI supervised several undergraduate research projects, participated in Vakil's Algebraic Geometry in the Time fo Co project. They published 8 new papers and preprints, which have appeared in top journals such as Algebra and Number Theory, Geometry and Topology, and Journal of Algebra.

Summary Statement

Overall, this is an impressive and ambitious proposal. The outlined research would greatly generalize work on certain moduli spaces. The outlined broader impacts are likely to have a str positive influence on inclusivity in the math research community.

This proposal falls in the middle third of ones I reviewed.

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